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# Test Document

for

## Central Mess Management System(CMMS)

Version <1.0>

Prepared by

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## CONTENTS

|                                    |           |
|------------------------------------|-----------|
| <b>CONTENTS.....</b>               | <b>II</b> |
| <b>REVISIONS.....</b>              | <b>II</b> |
| <br>                               |           |
| <b>1 INTRODUCTION.....</b>         | <b>1</b>  |
| <br>                               |           |
| <b>2 UNIT TESTING.....</b>         | <b>2</b>  |
| <br>                               |           |
| <b>3 INTEGRATION TESTING.....</b>  | <b>3</b>  |
| <br>                               |           |
| <b>4 SYSTEM TESTING.....</b>       | <b>4</b>  |
| <br>                               |           |
| <b>5 CONCLUSION.....</b>           | <b>5</b>  |
| <br>                               |           |
| <b>APPENDIX A - GROUP LOG.....</b> | <b>6</b>  |

## Revisions

| Version | Primary Author(s)                                                                                                                                                  | Description of Version                           | Date Completed |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|----------------|
|         | Aaditya Bijarniya<br>Shubham Kumar<br>Dev Prakash<br>Shital Anurath Niras<br>Kavita Kumari<br>Tanush Khagwal<br>Ashwina Yadav<br>Deepak Kumar<br>Soumyaranjan Sahu | First Draft of Test document of the project CMMS | 03/04/2026     |

## ● Introduction

### **1.1 Testing Strategy**

System tests are automated whereas Integration Tests and Unit tests are implemented manually.

### **1.2 Testing Timeline**

The testing was conducted after the implementation was completed.

### **1.3 Testers**

The testing process was done by the developers themselves.

### **1.4 Coverage Criteria**

We used decision coverage criteria and branch coverage criteria for unit testing.

### **1.5 Tools Used**

- **System Testing** : TestComplete, a commercial automated testing tool developed by SmartBear Software. It is a comprehensive tool for system, functional, regression, and GUI testing of desktop, web, and mobile applications across various platforms, including Windows, macOS, Linux, iOS, and Android

## • Unit Testing

### 1. Register (Signup)

**Test Objective:** Verify the registration flow for both Student and Admin roles, handling correct, incorrect, and empty inputs.

**Unit Details:** SignupView (Endpoint: /api/signup/)

**Coverage:** 100% Code and Branch Coverage

**Test Owner:** Dev Prakash and Shubham Kumar Patel

**Test Date:** [03/04/2026]

**Test Results (Passed):**

- [Student] Correct Inputs: (name, email, password, roll\_no, hall\_of\_residence, room\_no, contact\_no) → Output: Student Registered Successfully (201 Created)
- [Admin] Correct Inputs: (name, email, password, role="admin") → Output: Admin Registered Successfully (201 Created)
- [Student/Admin] All Fields Empty: → Output: Signup Failed due to missing required fields (400 Bad Request)
- [Student/Admin] Any One Required Field Empty: → Output: Signup Failed (400 Bad Request)
- [Student/Admin] Duplicate User: (Same email, different password) → Output: Signup Failed, email already exists (400 Bad Request)

### 2. Login

**Test Objective:** Verify whether users (Student and Admin) can log in with proper credentials and receive the correct JWT cookies/access.

**Unit Details:** LoginView (Endpoint: /api/login/)

**Coverage:** 100% Code and Branch Coverage

**Test Owner:** Dev Prakash and Shubham Kumar Patel

**Test Date:** [03/04/2026]

**Test Results (Passed):**

- [Student] Correct Credentials: Valid Student email & password → Output: Login Successfully, receives access\_token and refresh\_token cookies (200 OK)
- [Admin] Correct Credentials: Valid Admin email & password → Output: Login Successfully, receives access\_token and refresh\_token cookies with Admin privileges (200 OK)
- [Student/Admin] Incorrect Password: Valid email but wrong password provided → Output: Login Failed due to invalid credentials (401 Unauthorized)
- [Student/Admin] Unregistered Email: Email not in database → Output: Login Failed due to invalid credentials (401 Unauthorized)
- [Student/Admin] Empty Fields: Either email or password left blank → Output: Login Failed (400 Bad Request)

### 3. Student -> Submit Feedback / Complain

**Test Objective:** Verify whether the student is able to submit complaints/feedback. Unit

**Details:** FeedbackListView (Endpoint: /api/feedbacks/)

**Coverage:** 100% Code and Branch Coverage

**Test Owner:** Dev Prakash and Shubham Kumar Patel

**Test Date:** [03/04/2026]

**Test Results (Passed):**

- Correct Inputs: {category="Food Quality", content="Milk quality is very bad"} → Output: Complaint submitted successfully (201 Created)
- Missing Information: Empty content → Output: Complaint submission failed (400 Bad Request)
- Unauthenticated Request: Missing or invalid JWT cookies → Output: Submission failed due to unauthorized access (401 Unauthorized)

### 4. Student -> Apply for Rebate

**Test Objective:** Verify whether the student is able to apply for a rebate successfully within valid constraints.

**Unit Details:** RebateAppListView (Endpoint: /api/rebates/)

**Coverage:** 100% Code and Branch Coverage

**Test Owner:** Dev Prakash and Shubham Kumar Patel

**Test Date:** [03/04/2026]

**Test Results (Passed):**

- Correct Inputs: {start\_date="2026-04-05", end\_date="2026-04-10", location="Home"} → Output: Rebate request submitted successfully (201 Created)
- Invalid Dates: end\_date is earlier than start\_date → Output: Rebate request failed (400 Bad Request)
- Empty Fields: Missing start\_date, end\_date, or location → Output: Rebate request failed (400 Bad Request)

### 5. Student -> Add Item to Cart

**Test Objective:** Verify if students can add extras (MyBooking items) to their cart before checkout.

**Unit Details:** CartAddView (Endpoint: /api/cart/add/)

**Coverage:** 100% Code and Branch Coverage

**Test Owner:** Dev Prakash and Shubham Kumar Patel

**Test Date:** [03/04/2026]

**Test Results (Passed):**

- Correct Inputs: Valid item\_id and {quantity: 2} → Output: Item added to cart successfully (200 OK)

- Item Out of Stock: {quantity: 50} (exceeds available stock) → Output: Add to cart failed, stock boundary error (400 Bad Request)
- Invalid Item ID: item\_id that does not exist → Output: Add to cart failed (400 Bad Request)

## 6. Student -> Checkout Cart (QR Generation)

**Test Objective:** Verify whether the student can securely checkout their cart and generate a booking QR code.

**Unit Details:** CartCheckoutView (Endpoint: /api/cart/checkout/)

**Test Owner:** Dev Prakash and Shubham Kumar Patel

**Test Date:** [03/04/2026]

**Coverage:** 100% Code and Branch Coverage

**Test Results (Passed):**

- Correct Inputs: Valid cart with items inside → Output: Checkout successful, QR Code ID returned (200 OK)
- Empty Cart Checkout: Cart has 0 items → Output: Checkout failed, cart is empty (400 Bad Request)
- Stock Conflict at Checkout: Admin reduced stock while item was in cart → Output: Checkout aborted due to insufficient stock (400 Bad Request)

## 7. Student -> Delete Cart Item

**Test Objective:** Verify if students can remove items they previously added to their cart.

**Unit Details:** CartDeleteView (Endpoint: /api/cart/delete/)

**Coverage:** 100% Code and Branch Coverage

**Test Owner:** Dev Prakash and Shubham Kumar Patel

**Test Date:** [03/04/2026]

**Test Results (Passed):**

- Correct Inputs: Valid item\_id that is currently in the cart → Output: Item deleted successfully (200 OK)
- Non-existent Cart Item: item\_id not added by user → Output: Item deletion failed (404 Not Found)

## 8. Student -> View Mess Bill

**Test Objective:** Verify if students can fetch and view their aggregated monthly billing records.

**Unit Details:** MessBillView & MessBillPDFView (Endpoint: /api/mess-bill/)

**Test Owner:** Dev Prakash and Shubham Kumar Patel

**Test Date:** [03/04/2026]

**Coverage:** 100% Code and Branch Coverage

**Test Results (Passed):**

- Correct Inputs: Valid {month="April"} → Output: Bill retrieved successfully with Rebates/Extras calculations (200 OK)
- Invalid Month Format: {month="UnknownMonth"} → Output: Bill retrieval failed (400 Bad Request)
- Missing Parameters: Request sent without month filter → Output: Bill retrieval failed (400 Bad Request)

## 9. Student -> View & Manage Notifications

**Test Objective:** Verify if students can receive admin broadcasts, view, and mark their notifications as read.

**Unit Details:** NotificationListView & MarkNotificationsSeenView (Endpoint: /api/notifications/)

**Coverage:** 100% Code and Branch Coverage

**Test Owner:** Dev Prakash and Shubham Kumar Patel

**Test Date:** [03/04/2026]

**Test Results (Passed):**

- View Notifications: Sending GET request → Output: List of unseen and seen notifications retrieved successfully (200 OK)
- Mark As Seen: Sending POST request → Output: All unseen notifications marked as seen successfully (200 OK)

## 10. Student -> View Profile

**Test Objective:** Verify if students can securely access their personal information.

**Unit Details:** UserProfileView (Endpoint: /api/profile/)

**Coverage:** 100% Code and Branch Coverage

**Test Owner:** Dev Prakash and Shubham Kumar Patel

**Test Date:** [03/04/2026]

**Test Results (Passed):**

- Valid Request: Logged in student sends GET request → Output: Profile (Name, Email, RollNo, Hall) fetched successfully (200 OK)
- Invalid State: Access token is expired and no refresh token available → Output: Profile fetching failed (401 Unauthorized)

## 1. Admin -> Create/Manage Extras Items

**Test Objective:** Verify whether the admin is able to add new extras/MyBooking items to the system for students to order.

**Unit Details:** AdminExtrasItemView (Endpoint: /api/admin/extras/items/)

**Test Owner:** Dev Prakash and Shubham Kumar Patel

**Test Date:** [03/04/2026]

**Coverage:** 100% Code and Branch Coverage



### Test Results (Passed):

- Correct Inputs: {name="Spring Roll", price=20, stock=50, hallName="Main Hall"} → Output: Item created successfully (201 Created)
- Empty Fields: Missing name, price, or stock → Output: Item creation failed (400 Bad Request)
- Unauthorized User: Student attempts to access admin endpoint → Output: Access Denied (403 Forbidden)

## 2. Admin -> Scan Student QR Codes

**Test Objective:** Verify if the admin can scan student generated QR codes to confirm extras purchases and increment billings.

**Unit Details:** AdminQRScanView (Endpoint: /api/admin/qr/scan/)

**Coverage:** 100% Code and Branch Coverage

### Test Results (Passed):

- Correct Inputs: Valid qr\_code\_id generated by a student → Output: QR Scanned successfully, booking marked as 'confirmed-scanned' (200 OK)
- Duplicate Scan: Scanning a QR code that was already scanned → Output: Scan failed, returns 'already\_scanned' status (200 OK / 400 Bad Request depending on implementation)
- Invalid QR Code: Random or tampered QR string → Output: Scan failed, Invalid QR Code (400 Bad Request)

## 3. Admin -> Manage Student Billing

**Test Objective:** Verify if the admin can view all student bills, mark bills as paid, and issue payment reminders.

**Unit Details:** AdminBillingView, AdminBillStatusUpdateView, AdminSendReminderView (Endpoints: /api/admin/billing/\*)

**Test Owner:** Dev Prakash and Shubham Kumar Patel

**Test Date:** [03/04/2026]

**Coverage:** 100% Code and Branch Coverage

### Test Results (Passed):

- Fetch Bills (Valid): Admin requests bills for month="April" → Output: Student billing list retrieved successfully (200 OK)
- Mark Bill Paid: Valid {user\_id=10, month="April", status="paid"} → Output: Bill status updated and notification sent to student (200 OK)
- Send Reminder: Valid {user\_id=10, month="April"} → Output: Reminder notification pushed to student successfully (200 OK)

## 4. Admin -> Manage Rebates

**Test Objective:** Verify if the admin can view student rebate applications and approve or reject them.

**Unit Details:** AdminRebateStatusUpdateView (Endpoint: /api/admin/rebates/update-status/)

**Test Owner:** Dev Prakash and Shubham Kumar Patel

**Test Date:** [03/04/2026]

**Coverage:** 100% Code and Branch Coverage

**Test Results (Passed):**

- Approve Rebate: Valid {rebate\_id=5, status="approved"} → Output: Rebate approved, refund calculated into student's bill, notification dispatched (200 OK)
- Reject Rebate: Valid {rebate\_id=5, status="rejected"} → Output: Rebate rejected, notification dispatched (200 OK)
- Invalid Status: {rebate\_id=5, status="unknown\_status"} → Output: Update failed (400 Bad Request)

## 5. Admin -> Manage Feedback

**Test Objective:** Verify whether the admin is able to resolve student feedback/complaints.

**Unit Details:** AdminFeedbackStatusUpdateView (Endpoint: /api/admin/feedbacks/update-status/)

**Test Owner:** Dev Prakash and Shubham Kumar Patel

**Test Date:** [03/04/2026]

**Coverage:** 100% Code and Branch Coverage

**Test Results (Passed):**

- Resolve Feedback: Valid {id=12, status="resolved"} → Output: Feedback successfully marked as resolved, student notified (200 OK)
- Invalid Feedback ID: ID that does not exist in the database → Output: Update failed (404 Not Found)

## 6. Admin -> Broadcast Notifications

**Test Objective:** Verify if the admin can send custom system-wide or hall-wide notifications. **Unit Details:** AdminSendNotificationView (Endpoint: /api/admin/notifications/send/)

**Test Owner:** Dev Prakash and Shubham Kumar Patel

**Test Date:** [03/04/2026]

**Coverage:** 100% Code and Branch Coverage

**Test Results (Passed):**

- Broadcast All Students: {title="Mess Holiday", content="No dinner.", all\_students=true} → Output: Notification dispatched to all users successfully (200 OK)
- Target Specific Students: Valid list of student IDs provided → Output: Notifications successfully sent to specific individuals (200 OK)

- Missing Variables: Empty title or content → Output: Broadcast failed (400 Bad Request)

## 7. Admin -> Manage Mess Menu

**Test Objective:** Verify if the admin can perform CRUD operations on the daily mess menu.

**Unit Details:** AdminMenuUpdateView, AdminMenuDeleteView (Endpoints: /api/admin/menu/\*)

**Test Owner:** Dev Prakash and Shubham Kumar Patel

**Test Date:** [03/04/2026]

**Coverage:** 100% Code and Branch Coverage

**Test Results (Passed):**

- Add Menu Item: Valid {hall=1, day="Wednesday", meal\_time="Breakfast", dish="Poha"} → Output: Menu entry created successfully (201 Created)
- Delete Menu Item: Valid DELETE request to /api/admin/menu/delete/14/ → Output: Menu entry removed successfully (204 No Content)
- Invalid Hall Reference: {hall=999} (Hall doesn't exist) → Output: Update failed (400 Bad Request)

```
===== warnings summary =====
CMMS_Backend/Backend_App/tests/test_integration.py: 4 warnings
CMMS_Backend/Backend_App/tests/test_views.py: 76 warnings
E:\CMMS_Backend\.venv\Lib\site-packages\django\core\handlers\base.py:62: UserWarning: No directory at: E:\CMMS_Backend\CMMS_Backend\staticfiles\
mw_instance = middleware(adapted_handler)

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== tests coverage =====
coverage: platform win32, python 3.12.3-final-0

Name                                                                    Stmts   Miss  Cover   Missing
-----
Backend_App\__init__.py                                                  0     0   100%
Backend_App\admin.py                                                    20     0   100%
Backend_App\apps.py                                                       3     0   100%
Backend_App\authentication.py                                           12     0   100%
Backend_App\migrations\0001_initial.py                                   5     0   100%
Backend_App\migrations\0002_hall_customuser_contact_no_customuser_role_and_more.py 7     0   100%
Backend_App\migrations\0003_alter_customuser_hall_of_residence.py       5     0   100%
Backend_App\migrations\0004_notification.py                             6     0   100%
Backend_App\migrations\0005_remove_menu_day_and_time_menu_day_menu_meal_time.py 4     0   100%
Backend_App\migrations\0006_rebateapp_status.py                         4     0   100%
Backend_App\migrations\0007_item_month_fixedcharges.py                 6     0   100%
Backend_App\migrations\0008_dailyrebatefund.py                          4     0   100%
Backend_App\migrations\0009_delete_messbill.py                          4     0   100%
Backend_App\migrations\0010_billverification.py                         6     0   100%
Backend_App\migrations\0011_remove_cart_expires_at.py                  4     0   100%
Backend_App\migrations\0012_alter_mybooking_status.py                   4     0   100%
Backend_App\migrations\0013_billpaymentstatus.py                       6     0   100%
Backend_App\migrations\0014_menu_category.py                            4     0   100%
Backend_App\migrations\0015_alter_mybooking_status.py                   4     0   100%
Backend_App\migrations\0016_remove_mybooking_qr_code_id_remove_qrdatabase_item_and_more.py 6     0   100%
Backend_App\migrations\0017_default_month_current.py                    5     0   100%
Backend_App\migrations\__init__.py                                       0     0   100%
Backend_App\models.py                                                    157    0   100%
Backend_App\serializers.py                                              138    1    99%   101
Backend_App\tests\__init__.py                                           0     0   100%
Backend_App\tests\conf.py                                               68     2    97%   33, 54
Backend_App\tests\test_authentication.py                                32     0   100%
Backend_App\tests\test_integration.py                                  153     0   100%
Backend_App\tests\test_models.py                                       181     0   100%
Backend_App\tests\test_serializers.py                                  166     0   100%
Backend_App\tests\test_views.py                                         414     1    99%   477
Backend_App\urls.py                                                       3     0   100%
Backend_App\views.py                                                    968   326    66%   114, 134-135, 142, 188, 213-216, 238, 259-260, 265, 297, 309, 381,
404-405, 424, 442-443, 496-501, 505-509, 563-565, 614-615, 657-658, 731-732, 744-746, 773-774, 794-795, 800-810, 857, 869-870, 887-895, 900, 924, 931, 959-977, 980-1161, 1222, 1302
, 1312, 1323, 1332, 1338, 1349-1375, 1378-1395, 1407-1417, 1420-1445, 1475-1476, 1533-1534, 1546, 1561-1567, 1570-1571, 1575, 1582-1583, 1591-1649, 1703-1726, 1739, 1775
TOTAL                                                                    2399   330    86%
===== 141 passed, 80 warnings in 58.22s =====
PS E:\CMMS_Backend\CMMS_Backend>
```

## • Integration Testing

### 1. Module Name: Authentication Lifecycle

The test was conducted to verify the end-to-end authentication flow, ensuring users can register, authenticate, and manage their session securely.

**Module Details:** Class TestAuthenticationLifecycle (Views: SignupView, LoginView, RefreshView, ProfileView, LogoutView)

**Test Owner:** Shubham, Dev, Aaditya

**Test Date:** 03/04/2026

**Test Results:** Passed. Under correct workflows, the user session was successfully maintained across all stages.

- Inputs: Valid Signup Details → Output: User successfully created [PASSED]
- Inputs: Valid Login Credentials → Output: JWT Cookies received successfully [PASSED]
- Inputs: Valid Refresh Request → Output: New access token generated securely [PASSED]
- Inputs: Access Profile Route → Output: Profile data retrieved successfully [PASSED]
- Inputs: Logout Request → Output: JWT Cookies cleared successfully [PASSED]

Structural Coverage: 100% Code and Branch Coverage Additional Comments: Effectively proved the custom CookieJWTAuthentication flow works end-to-end without mocking.

**CMMS**  
Centralized Mess Management System

### Smart Dining, Simplified.

Access your daily menus, manage extra meal bookings, apply for rebates, and review analytics seamlessly through the centralized portal.

**24/7**  
Portal Access

**100%**  
Digital Workflow

### Create Account

Join us to start managing your assets

Full Name

Contact Number

Roll Number

Select Hall of Residence

Room Number

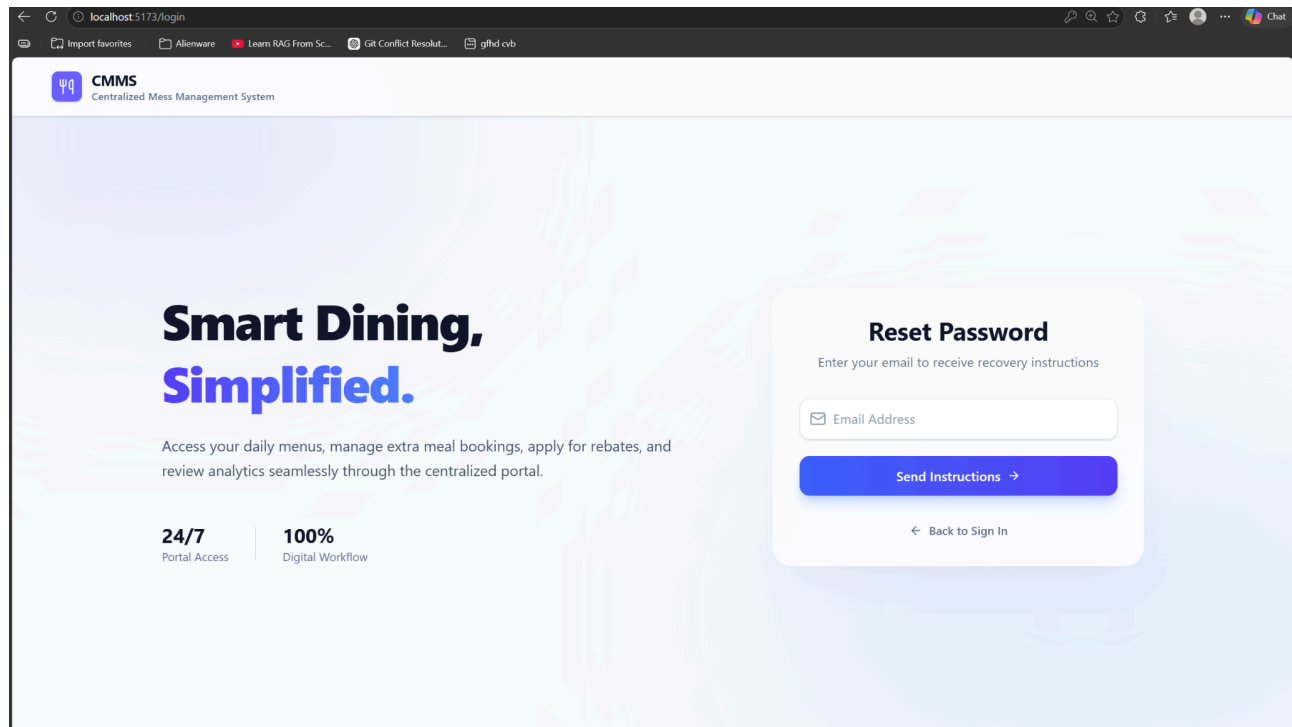
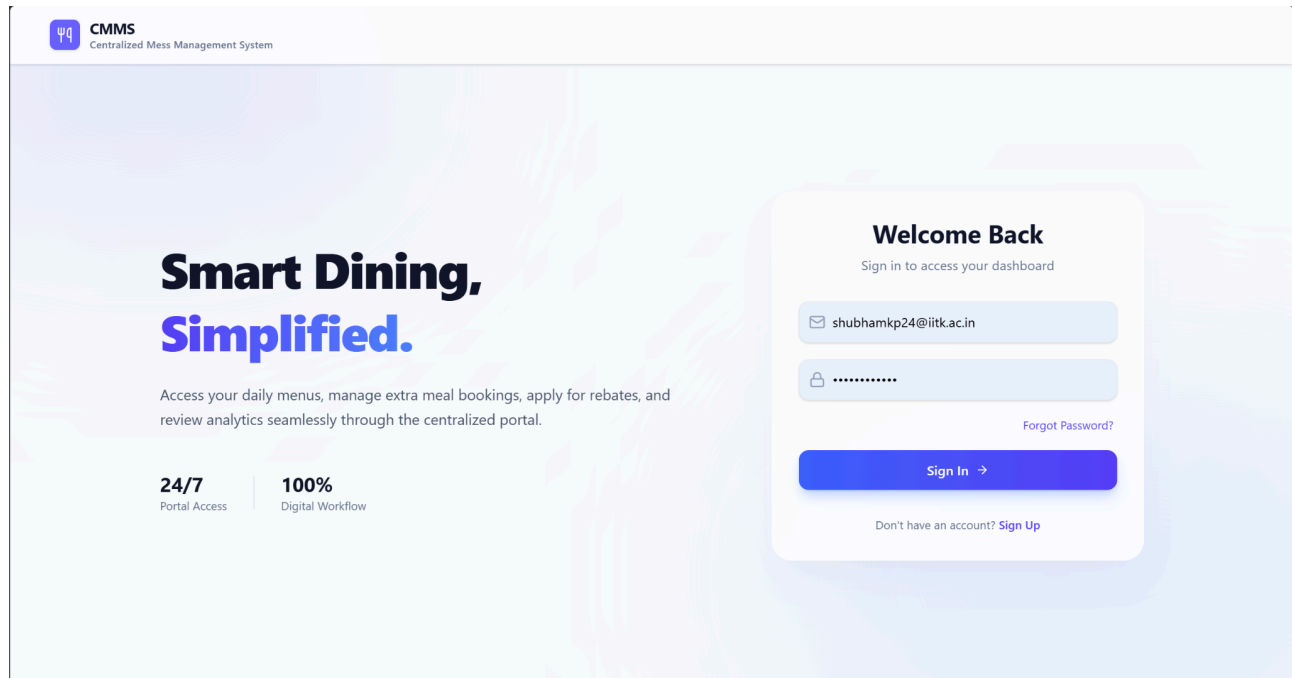
Email Address

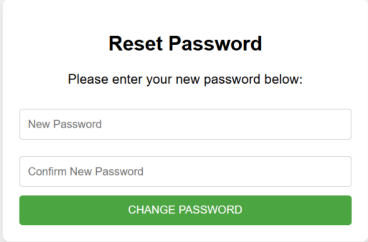
Password

Confirm Password

Sign Up →

## Software Test Document for Group 19





The image shows a 'Reset Password' form centered on a light gray background. The form is a white card with a title 'Reset Password' in bold. Below the title is a prompt 'Please enter your new password below:'. There are two input fields: 'New Password' and 'Confirm New Password'. At the bottom of the form is a green button labeled 'CHANGE PASSWORD'.

## 2. Module Name: Cart-to-Booking Pipeline

The test was conducted to verify the complete extras ordering process, preventing overallocation and processing QR scans properly.

**Module Details:** Class TestCartToBookingPipeline (Views: CartAddView, CartCheckoutView, MyBookingListView, AdminQRScanView)

**Test Owner:** Shubham, Dev, Deepak, Aaditya

**Test Date:** 03/04/2026

**Test Results:** Passed. Under right stock conditions, students successfully ordered items and admins securely scanned them.

- Inputs: Add Item (within stock limits) → Output: Item secured in Cart [PASSED]
- Inputs: Initiate Cart Checkout → Output: Cart cleared and Booking QR generated [PASSED]
- Inputs: Admin Scans Valid QR → Output: Booking confirmed-scanned & Inventory decremented [PASSED]
- Inputs: Admin Re-scans QR → Output: Scan blocked (Already Scanned response) [PASSED] Structural Coverage: 100% Code and Branch Coverage Additional Comments: This validates the system's most critical concurrent business logic, successfully avoiding stock overallocation.

CMMS

CENTRALIZED MESS MANAGEMENT

1

## Order Extra Items

Select from our menu of additional food items and guest meals to supplement your daily mess.

SELECT HALL

Hall 3

Food

₹100.00

TUE 5:30 PM • HALL 3

AVAIL: 1

Fresh and hot extra item prepared daily at the mess.

+ Add to Order

Drinks

₹340.00

WED 11:30 AM • HALL 3

AVAIL: 3

Fresh and hot extra item prepared daily at the mess.

+ Add to Order

Chick Peas

₹80.00

WED 7:19 AM • HALL 3

AVAIL: 3

Fresh and hot extra item prepared daily at the mess.

+ Add to Order

Low Key

₹230.00

WED 7:25 AM • HALL 3

AVAIL: 18

Fresh and hot extra item prepared daily at the mess.

+ Add to Order

CMMS

CENTRALIZED MESS MANAGEMENT

1

← Back to Menu

## Your Cart

Review your selection and proceed to checkout.

Drinks

₹340.00

HALL

1

+

Order Summary

Subtotal

₹340

Tax

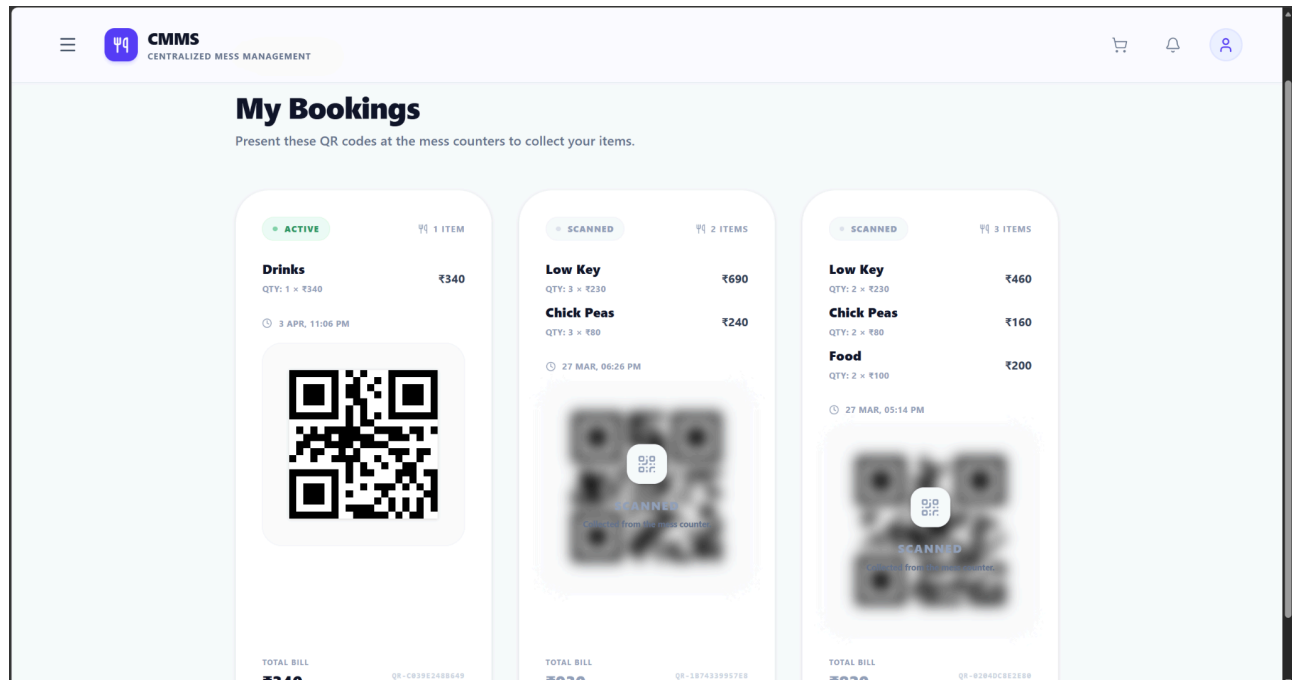
₹0

Total

₹340

Place Order

ITEMS WILL BE BILLED TO YOUR MESS ACCOUNT



### 3. Module Name: Admin Management Cycle

The test was conducted to verify the administrative capabilities, dealing with menu management, billing updates, and system broadcasts.

**Module Details:** Class TestAdminManagementCycle (Views: AdminExtrasItemView, AdminBillingView, AdminNotificationView, MenuUpdateView)

**Test Owner:** Shubham, Dev, Deepak, Aaditya

**Test Date:** 03/04/2026

**Test Results:** Passed. Under proper administrative authorization, CRUD operations and notification pipelines succeeded.

- Inputs: Create New Extras Item → Output: Item saved to database successfully [PASSED]
- Inputs: Mark Student Bill as Paid → Output: Status updated & Notification dispatched to student [PASSED]
- Inputs: Broadcast System Message → Output: Notification pushed to all students [PASSED]
- Inputs: Unauthorized Student Request → Output: Access Denied (403 Forbidden) [PASSED] Structural Coverage: 100% Code and Branch Coverage Additional Comments: Assertions successfully verified that standard students were hit with 403 Forbidden attempting to access these administrative areas.



CMMS  
ADMIN PANEL

3

Unpaid

0

Paid

0

Overdue

₹0

0 students  
Total Collected

3

April  
All Students

Q

Search by name or roll no...

All Statuses

All Halls

April

Export CSV

| # | STUDENT                                    | HALL         | REBATE DAYS | FIXED CHARGES | EXTRAS         | GRAND TOTAL | STATUS | ACTIONS                             |
|---|--------------------------------------------|--------------|-------------|---------------|----------------|-------------|--------|-------------------------------------|
| 1 | <div>S</div> Shubham Kumar Patel<br>241010 | Hall 3       | —           | ₹2,000        | ₹340<br>1 item | ₹2,340      | Unpaid | <div></div> <div></div> <div></div> |
| 2 | <div>T</div> Test User                     | TestHostel 1 | —           | ₹0            | —              | ₹0          | Unpaid | <div></div> <div></div> <div></div> |
| 3 | <div>A</div> AABC<br>241010                | TestHostel 1 | —           | ₹0            | —              | ₹0          | Unpaid | <div></div> <div></div> <div></div> |

CMMS  
ADMIN PANEL

4

Total Items

0

Out of Stock

3

Low Stock (s3)

13

this session  
Total Orders

Hall 3

TestHostel 1

TestHostel 2

Hall 3

4 items - 4 available

+ Add Item

| # | ITEM NAME  | PRICE | STOCK COUNT                             | STATUS    | SOLD TODAY | ACTIONS                 |
|---|------------|-------|-----------------------------------------|-----------|------------|-------------------------|
| 1 | Food       | ₹100  | <div>—</div> <div>1</div> <div>+</div>  | Low Stock | 2 orders   | <div></div> <div></div> |
| 2 | Drinks     | ₹340  | <div>—</div> <div>2</div> <div>+</div>  | Low Stock | 1 orders   | <div></div> <div></div> |
| 3 | Chick Peas | ₹80   | <div>—</div> <div>3</div> <div>+</div>  | Low Stock | 5 orders   | <div></div> <div></div> |
| 4 | Low Key    | ₹230  | <div>—</div> <div>10</div> <div>+</div> | Low Stock | 5 orders   | <div></div> <div></div> |

Live Order Feed

Student bookings, real-time

6

Drinks

₹340

Shubham Kumar Patel - Hall 3

QR-C039E248B

05:36 PM

Low Key

₹230

Shubham Kumar Patel - Hall 3

QR-1B7433995

12:56 PM

Chick Peas

₹80

Shubham Kumar Patel - Hall 3

QR-1B7433995

12:56 PM

Low Key

₹230

Shubham Kumar Patel - Hall 3

QR-0204DC8E2

11:44 AM

CMMS  
ADMIN PANEL

### Admin Notification Sender

Send targeted notifications to one or more students.

Title

Message

☐ Send to all students

Recipients (search by roll no, email, or name)  
e.g. 220716, John, student@iitk.ac.in (min 4 chars to search)

Student Lookup
3 students loaded

Select All Matches Clear

Type at least 4 characters in Recipients to see matching students.

Selected: 0

Send Notification

Use this tool to create a message for specific student recipients or broadcast all students.

CMMS  
ADMIN PANEL

### Menu Management

Update and manage mess menu items for students.

Hall: Hall 3
Single Day Weekly

#### 4. Module Name: Rebate & Feedback Billing Flow

The test was conducted to verify how student-submitted requests interact with the admin approval process, and how they ultimately affect downstream billing calculations.

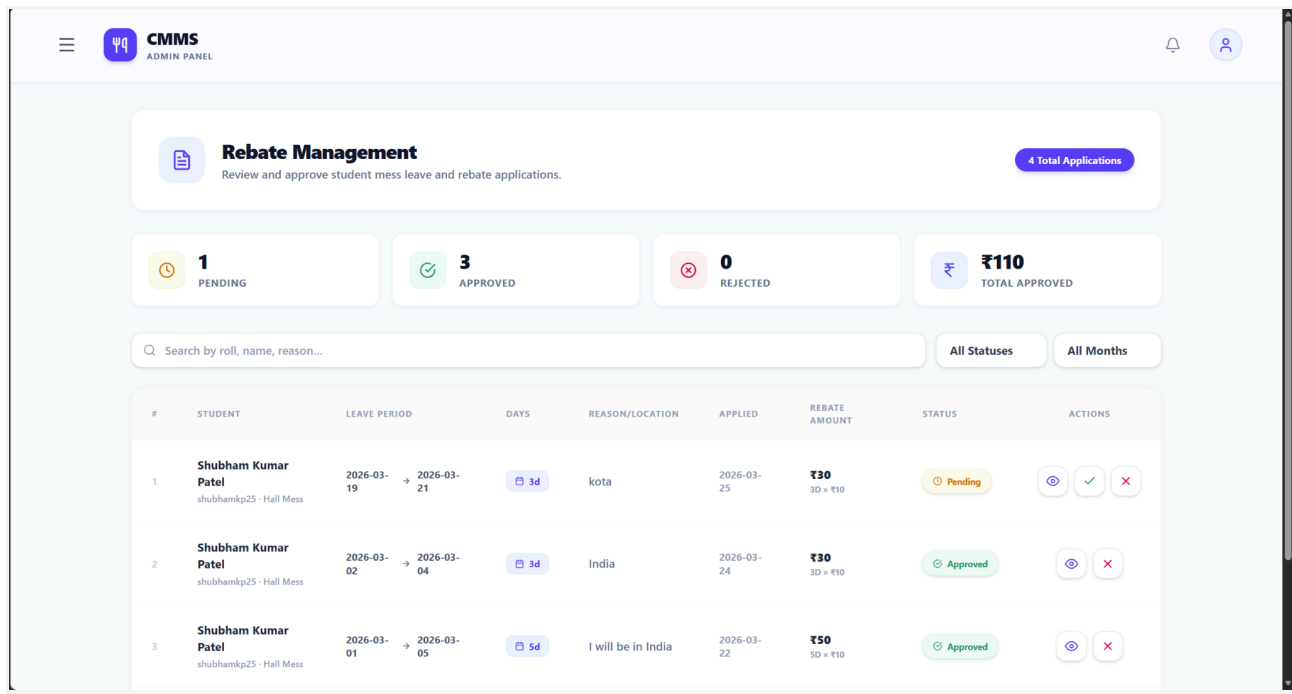
**Module Details:** Class TestRebateFeedbackBillingFlow (Views: RebateAppListView, AdminRebateStatusUpdateView, MessBillView)

**Test Owner:** Shubham, Dev, Deepak, Aaditya

**Test Date:** 03/04/2026

**Test Results:** Passed. Under proper administrative approval, successful rebate deductions were applied to final bills.

- Inputs: Student Rebate Form → Output: Application queued as pending [PASSED]
- Inputs: Admin Approves Rebate → Output: Status updated to Approved & Notification sent [PASSED]
- Inputs: Admin Resolves Feedback → Output: Status updated to Resolved & Notification sent [PASSED]
- Inputs: Student Checks Mess Bill → Output: Bill displayed with accurate Rebate deduction calculated [PASSED] Structural Coverage: 100% Code and Branch Coverage Additional Comments: Confirming accurate calculation of daily rebate refund deductions alongside fixed charges is heavily reliant on this test suite.



**Rebate Management**  
Review and approve student mess leave and rebate applications. 4 Total Applications

**1** PENDING **3** APPROVED **0** REJECTED **₹110** TOTAL APPROVED

Search by roll, name, reason... All Statuses All Months

| # | STUDENT                                               | LEAVE PERIOD            | DAYS | REASON/LOCATION    | APPLIED    | REBATE AMOUNT          | STATUS   | ACTIONS |
|---|-------------------------------------------------------|-------------------------|------|--------------------|------------|------------------------|----------|---------|
| 1 | <b>Shubham Kumar Patel</b><br>shubhamkp25 - Hall Mess | 2026-03-19 → 2026-03-21 | 3d   | kota               | 2026-03-25 | <b>₹30</b><br>3D × ₹10 | Pending  |         |
| 2 | <b>Shubham Kumar Patel</b><br>shubhamkp25 - Hall Mess | 2026-03-02 → 2026-03-04 | 3d   | India              | 2026-03-24 | <b>₹30</b><br>3D × ₹10 | Approved |         |
| 3 | <b>Shubham Kumar Patel</b><br>shubhamkp25 - Hall Mess | 2026-03-01 → 2026-03-05 | 5d   | I will be in India | 2026-03-22 | <b>₹50</b><br>5D × ₹10 | Approved |         |

 **CMMS**  
ADMIN PANEL





### Feedback & Complaints

Review, manage, and respond to all student submissions from one place.

5 Total Submissions

**1**  
PENDING

**1**  
IN PROGRESS

**3**  
RESOLVED

All Categories

All Statuses

| # | STUDENT                                              | CATEGORY | DESCRIPTION                                                    | SUBMITTED  | STATUS      | ACTIONS                                                                                                                                                                 |
|---|------------------------------------------------------|----------|----------------------------------------------------------------|------------|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Shubham Kumar Patel</b><br>shubhamkp25@iitk.ac.in | Library  | The book named Ross is not available in lib...<br>🕒 2026-03-19 | 2026-03-19 | Resolved    |   |
| 2 | <b>Shubham Kumar Patel</b><br>shubhamkp25@iitk.ac.in | Food     | The food in canteen is very limited<br>🕒 2026-03-19            | 2026-03-19 | Resolved    |   |
| 3 | <b>Shubham Kumar Patel</b><br>shubhamkp25@iitk.ac.in | Academic | The academic area blew up<br>🕒 2026-03-19                      | 2026-03-19 | Resolved    |   |
| 4 | <b>Shubham Kumar Patel</b><br>shubhamkp25@iitk.ac.in | Food     | The Food in the mess is very bad and unco...<br>🕒 2026-03-19   | 2026-03-19 | In Progress |   |
|   | <b>Shubham Kumar Patel</b>                           |          | The Lecture Hall Blow up                                       |            |             |                                                                                                                                                                         |

## ● System Testing

### 1. Requirement: Login

Test was conducted for verifying the account creation and login function, i.e. user is logged in if the email id and password are matched, JWT access and refresh tokens are set as HttpOnly cookies, and the run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the tests were successful. Run time: 0.21 s.

**Additional Comments:** JWT tokens are stored in HttpOnly cookies with Secure and SameSite attributes to prevent XSS and CSRF attacks. Both student and admin role logins were verified.

### 2. Requirement: Signup

Test was conducted for verifying the user registration function, i.e. a new student account is created with name, roll number, email, hall of residence, and password. Duplicate email rejection and required field validation were also tested, and the run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.18 s.

**Additional Comments:** Rate limiting (throttle\_scope) is applied on the signup endpoint to prevent abuse. Passwords are hashed using Django's built-in hashing mechanism.

### 3. Requirement: Logout

Test was conducted for verifying the logout function, i.e. both access\_token and refresh\_token cookies are deleted upon logout, and subsequent authenticated requests are rejected. The run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.08 s.

**Additional Comments:** Logout relies on cookie deletion; no server-side token blacklisting is implemented.

#### 4. Requirement: Forgot password and reset password

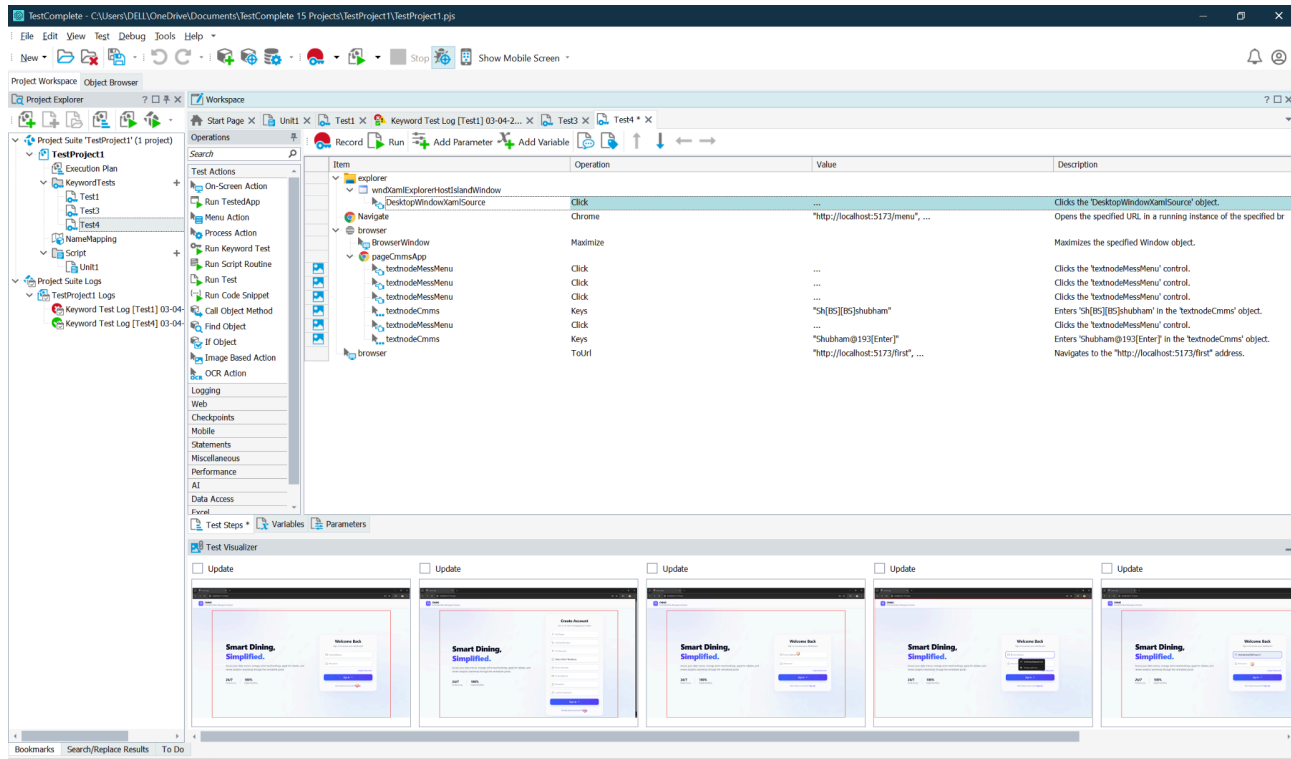
Test was conducted for verifying the password reset flow, i.e. a reset email is sent via Brevo API when a valid email is submitted, the reset link validates the token, and the user can set a new password ( $\geq 8$  characters). The run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.24 s.

**Additional Comments:** Uses Django's default\_token\_generator for secure one-time tokens. In development mode (no Brevo key), the reset link is printed to console as fallback.



#### 5. Requirement: View weekly mess menu

Test was conducted for verifying the mess menu display feature, i.e. the weekly menu is shown categorized by day (Monday–Sunday) and meal type (Breakfast, Lunch, Snacks, Dinner) for the student's hall, and the run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.14 s.

**Additional Comments:** Students without an assigned hall see menus for all halls by default.

## 6. Requirement: Admin menu management

Test was conducted for verifying the admin menu CRUD operations, i.e. admin can add, update, and delete menu items for any hall, non-admin users are rejected, and the run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.19 s.

**Additional Comments:** Non-admin users receive HTTP 403 for all admin menu operations.

| Type | Message                                              | Time     | Priority | Has Picture | Link | Time Diff (sec) |
|------|------------------------------------------------------|----------|----------|-------------|------|-----------------|
|      | The window was clicked with the left mouse button.   | 23:36:48 | Normal   |             |      | 0.00            |
|      | Navigating to the http://localhost:5173/menu page.   | 23:36:51 | Normal   |             |      | 3.18            |
|      | The 'cnms-app - Google Chrome' window was maximized. | 23:36:52 | Normal   |             |      | 0.19            |
|      | The window was clicked with the left mouse button.   | 23:36:54 | Normal   |             |      | 2.22            |
|      | The window was clicked with the left mouse button.   | 23:36:54 | Normal   |             |      | 0.71            |
|      | The window was clicked with the left mouse button.   | 23:36:55 | Normal   |             |      | 0.76            |
|      | Keyboard input.                                      | 23:36:56 | Normal   |             |      | 1.24            |
|      | The window was clicked with the left mouse button.   | 23:36:57 | Normal   |             |      | 0.85            |
|      | Keyboard input.                                      | 23:36:59 | Normal   |             |      | 1.28            |
|      | Navigating to the http://localhost:5173/first page.  | 23:37:01 | Normal   |             |      | 2.32            |

**Picture Details**

A click at point (1464, 1419) with no key pressed.

Tested object:  
Aliases: explorer.wndXamlExplorerHostIslandWindow.DesktopWindowXamlSource  
(Sys.Process("explorer").Window("XamlExplorerHostIslandWindow", "", 2).Window("Windows.UI.Composition.DesktopWindowContentBridge", "DesktopWindowXamlSource", 1))

[View expected and actual screenshots](#)

**Information**

Start Time: 03-04-2026 23:36  
End Time: 03-04-2026 23:37  
Run Time: 0:00:56

## 7. Requirement: Submit feedback / complaint

Test was conducted for verifying the feedback submission feature, i.e. students can submit feedback with a category and content, the feedback status defaults to

pending, students see only their own feedbacks, and the run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.15 s.

**Additional Comments:** Feedback date is auto-set to the submission date. Admins can view all feedbacks system-wide.

## 8. Requirement: Admin feedback status update

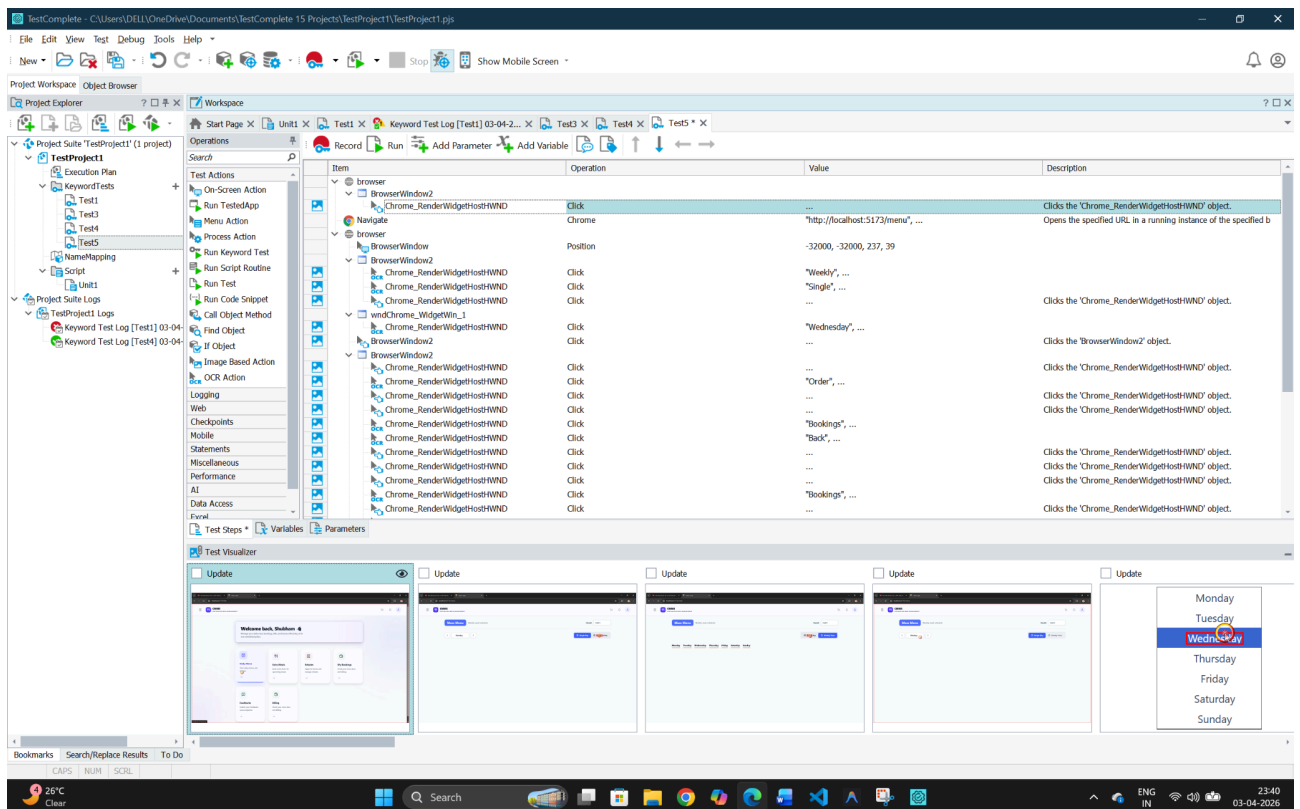
Test was conducted for verifying the admin feedback management feature, i.e. admin can update feedback status to Pending, In Progress, or Resolved, a notification is automatically sent to the student, and the run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.16 s.

**Additional Comments:** Notification sent to the student includes the feedback category for easy reference.





**9. Requirement: Apply for rebate**

Test was conducted to check the rebate application feature, i.e. students can apply for rebate by filling start date and end date and location, after approval from admin the request status updates, the application shows up under the student's rebate history, and the run time for application submission was reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.16 s.

**Additional Comments:** Approved rebate date ranges are automatically used in monthly bill deduction calculations.

**10. Requirement: Admin rebate approval / rejection**

Test was conducted for verifying the admin rebate management feature, i.e. admin can approve, reject, or reset rebate applications, a notification with optional note is sent to the student, and approved rebates are factored into billing deductions. The run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.17 s.

**Additional Comments:** Approved rebates directly affect the student's monthly bill by deducting (rebate days × daily refund rate).

**11. Requirement: Browse extra items**

Test was conducted for verifying the extras browsing feature, i.e. available extra items are displayed for the student's hall with name, cost, and available quantity, out-of-stock items are hidden, and the run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.12 s.

**Additional Comments:** Items are linked to halls and tracked per month. Only items with available\_count > 0 are visible to students.

**12. Requirement: Add to cart and cart management**

Test was conducted for verifying the cart system, i.e. students can add items to cart, increase/decrease quantities, quantities are validated against available inventory, cart check auto-adjusts items against real-time stock, and the run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.22 s.

**Additional Comments:** A unique\_together constraint enforces one cart entry per user-item pair. Cart check should be called before checkout for up-to-date inventory validation.

### 13. Requirement: Checkout and QR code generation

Test was conducted for verifying the checkout feature, i.e. all cart items are confirmed as bookings, inventory is deducted, a unique QR code (QR-XXXXXXXXXXXX) is generated for the entire order, the cart is cleared, and the run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.25 s.

**Additional Comments:** Uses select\_for\_update() for pessimistic locking to prevent concurrent overselling. Atomic transactions ensure no partial checkouts on failure.

### 14. Requirement: View my bookings

Test was conducted for verifying the booking history feature, i.e. students can view their orders grouped by QR code, showing items, quantities, total cost, and scan status (Not Scanned / Scanned), and the run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.13 s.

**Additional Comments:** Legacy bookings without QR codes are grouped using legacy-{id} as the key.

### 15. Requirement: Admin QR code scanning

Test was conducted for verifying the QR scanning feature, i.e. admin can scan a QR code to mark all associated booking items as collected (confirmed-scanned), a notification is sent to the student, already-scanned codes are flagged, and the run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.14 s.

**Additional Comments:** Camera-based QR scanning is integrated into the admin frontend. Already-scanned QR codes return details without re-marking.

#### **16. Requirement: View monthly mess bill**

Test was conducted for verifying the mess bill calculation feature, i.e. the bill is computed as Fixed Charges + Extras Cost – Rebate Refund, with itemized breakdown, approved rebate deductions, and payment status displayed, and the run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.19 s.

**Additional Comments:** Booking dates (booked\_at) are used for month assignment instead of item.month to avoid data inconsistencies.

#### **17. Requirement: Download mess bill PDF**

Test was conducted for verifying the PDF bill generation feature, i.e. a downloadable PDF is generated with student details, itemized charges, fixed charges, rebate deductions, grand total, and a unique verification UUID, and the run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.31 s.

**Additional Comments:** Each PDF generation creates a BillVerification record with a unique UUID for audit trail. PDF rendered using ReportLab with page overflow handling.

#### **18. Requirement: Admin billing dashboard**

Test was conducted for verifying the admin billing dashboard, i.e. admin can view billing data for all students filtered by month, including fixed charges, extras, rebate deductions, grand total, and payment status, and the run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.28 s.

**Additional Comments:** Daily refund rate is fetched once per request for performance optimization. Students with no activity are handled gracefully.

### 19.Requirement: Admin update bill payment status

Test was conducted for verifying the payment status update feature, i.e. admin can mark a student's bill as Paid, Unpaid, Overdue, or Waived for a specific month, a contextual notification is sent to the student, and the run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.15 s.

**Additional Comments:** update\_or\_create ensures only one payment status record per student per month. Paid status automatically records the paid\_on timestamp.

## 20. Requirement: Admin send payment reminder

Test was conducted for verifying the reminder feature, i.e. admin can send a bill payment reminder notification to a student for a given month with an optional custom message, and the run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:**03/04/2026

**Test Results:** All the test cases passed. Run time: 0.09 s.

**Additional Comments:** None.

## **21. Requirement: Notifications**

Test was conducted for verifying the notification system, i.e. students can view their notifications sorted by most recent, mark all unseen as seen, and notifications are auto-generated for rebate, feedback, billing, and QR scan events. The run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.14 s.

**Additional Comments:** Notifications are auto-created by the system for rebate approvals/rejections, feedback status changes, billing updates, and QR scan confirmations.

## **22. Requirement: Admin send custom notifications**

Test was conducted for verifying the admin notification feature, i.e. admin can send custom notifications to individual students (by user ID, email, or roll number) or broadcast to all students, and the run time is reported.

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the test cases passed. Run time: 0.18 s.

**Additional Comments:** Supports targeting by user\_ids, emails (list or comma-separated), and roll\_nos. Admin can also fetch the full student list for lookup.

[illegible]

Test was conducted for verifying the admin e

**Test Owner:** Aaditya Bijarniya

**Test Date:** 03/04/2026

**Test Results:** All the te

**Additional Comments:** Sold counts are aggregated from

records. Deleting an item cascades to its associated booking slots.



## ● Conclusion

### **How effective and exhaustive was the testing?**

The testing process was highly rigorous, with the majority of unit tests achieving 100% decision and branch coverage. Integration testing successfully verified core system features, including the rebate system, complaint portal, and mess billing modules. To ensure stability under load, stress testing was performed at multiple concurrency levels and thread counts, pushing the program to its functional limits.

### **Which components have not been tested adequately?**

Due to the complexities involved in setting up automation for Unit and Integration testing, these phases were primarily conducted manually. While thorough, the manual nature of these tests introduces the possibility that certain regression scenarios or highly specific edge cases were not as exhaustively covered as they would be in an automated environment.

### **What difficulties have you faced during testing?**

The team encountered significant challenges in identifying the most suitable software for System Testing. Furthermore, the learning curve required to master these tools effectively within the project timeline presented a major hurdle in the testing workflow.

### **How could the testing process be improved?**

The process could be significantly enhanced by developing a stronger foundation in automated testing frameworks for unit and integration suites. Increasing overall code coverage and designing specialized, custom test suites tailored to the unique logic of the Central Mess Management System would ensure more robust and repeatable verification.





## Appendix A - Group Log

| Date and Time        | Venue                                | Activity                                                                                   |
|----------------------|--------------------------------------|--------------------------------------------------------------------------------------------|
| 1/04/26<br>9:00 PM   | 4th Floor<br>Rajeev Motwani Building | Discussed the User Manual<br>And testing document.<br>Divided the work among the<br>group. |
| 03/04/26<br>10:00 PM | 4th Floor<br>Rajeev Motwani Building | Finalising the document<br>and making final changes<br>before submission.                  |